

EXPERIMENT 9

AIM: Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.

SOLUTION:

A **stored function** is a named PL/SQL block that:

- ✓ Always **returns a value**

Syntax of Creating a Function

```
CREATE OR REPLACE FUNCTION function_name(  
    parameter_name datatype  
)  
RETURN datatype  
IS  
BEGIN  
    statements;  
    RETURN value;  
END;  
/
```

Create table:

```
CREATE TABLE student_fn (  
    rollno NUMBER PRIMARY KEY,  
    name VARCHAR2(20),  
    marks NUMBER  
);  
  
INSERT INTO student_fn VALUES (1,'Ravi',78);  
INSERT INTO student_fn VALUES (2,'Anu',45);  
INSERT INTO student_fn VALUES (3,'Kiran',90);  
  
COMMIT;
```

Create stored function:

```
CREATE OR REPLACE FUNCTION student_grade(p_roll NUMBER)  
RETURN VARCHAR2  
IS  
    v_marks NUMBER;  
    v_grade VARCHAR2(20);  
BEGIN  
    SELECT marks INTO v_marks  
    FROM student_fn
```

```

WHERE rollno = p_roll;

IF v_marks >= 75 THEN

    v_grade := 'DISTINCTION';

ELSIF v_marks >= 35 THEN

    v_grade := 'PASS';

ELSE

    v_grade := 'FAIL';

END IF;

RETURN v_grade;

EXCEPTION

    WHEN NO_DATA_FOUND THEN

        RETURN 'INVALID ROLL NO';

END;

/

```

Call function in SQL:

```

SELECT rollno, name, marks, student_grade(rollno) AS grade
FROM student_fn;

```

Output:

Rollno	Name	Marks	Grade
1	Ravi	78	DISTINCTION
2	Anu	45	PASS
3	Kiran	90	DISTINCTION